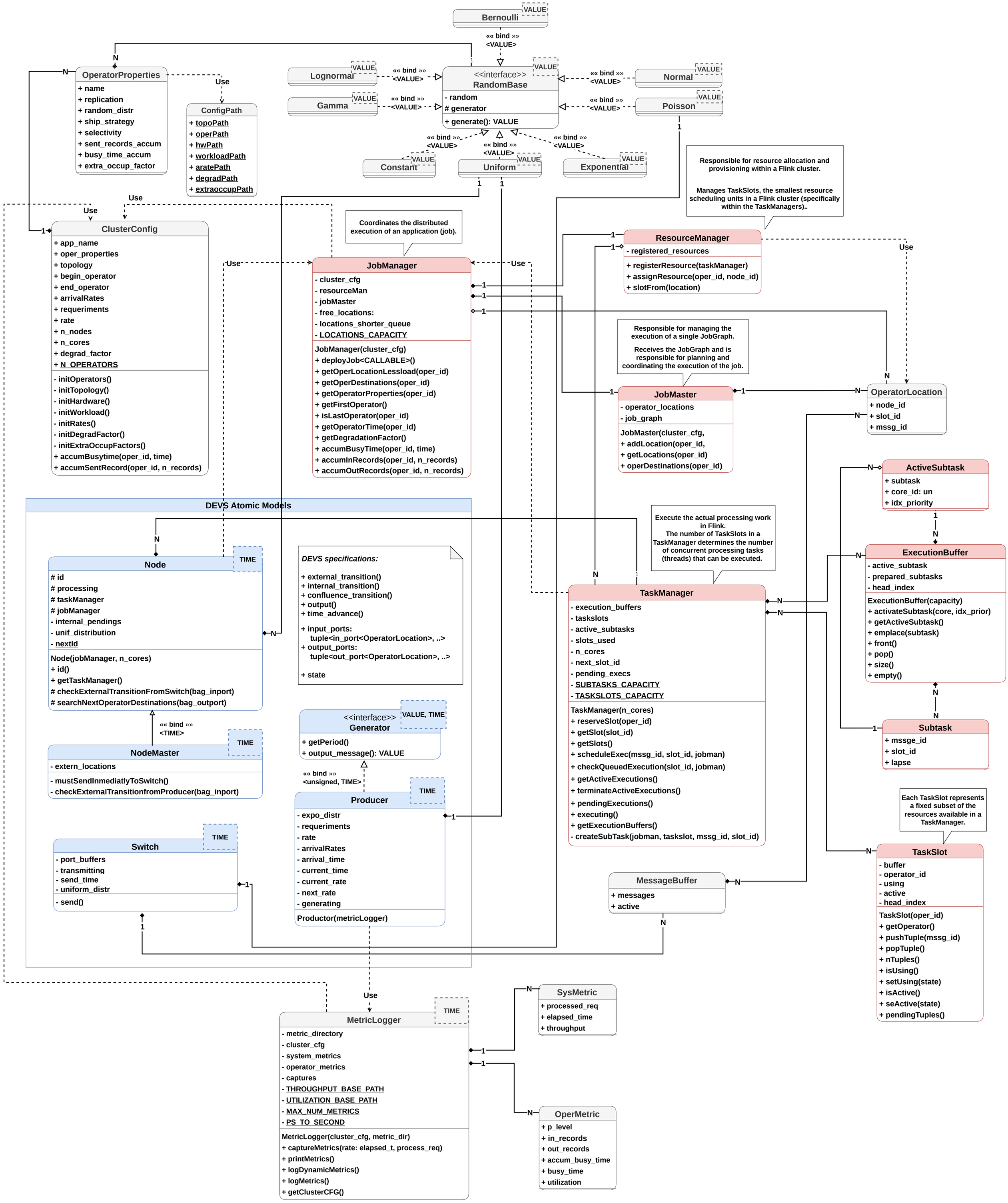

Bernoulli
VALUE
«« bind »»
<VALUE>
N
1
OperatorProperties
+ name
+ replication
+ random_distr
+ ship_strategy
+ selectivity
+ sent_records_accum
+ busy_time_accum
+ extra_occup_factor
Use
ConfigPath
+ topoPath
+ operPath
+ hwPath
+ workloadPath
+ aratePath
+ degradPath
+ extraoccupPath
Lognormal
Gamma
<<interface>>
RandomBase
- random
generator
+ generate(): VALUE
Normal
Poisson
Constant
Uniform
Exponential
Responsible for resource allocation and provisioning within a Flink cluster.
Manages TaskSlots, the smallest resource scheduling units in a Flink cluster (specifically within the TaskManagers)..
ClusterConfig
+ app_name
+ oper_properties
+ topology
+ begin_operator
+ end_operator
+ arrivalRates
+ requeriments
+ rate
+ n_nodes
+ n_cores
+ degrad_factor
+ N_OPERATORS
- initOperators()
- initTopology()
- initHardware()
- initWorkload()
- initRates()
- initDegradFactor()
- initExtraOccupFactors()
+ accumBusytime(oper_id, time)
+ accumSentRecord(oper_id, n_records)
Coordinates the distributed execution of an application (job).
JobManager
- cluster_cfg
- resourceMan
- jobMaster
- free_locations:
- locations_shorter_queue
- LOCATIONS_CAPACITY
JobManager(cluster_cfg)
+ deployJob<CALLABLE>()
+ getOperLocationLessload(oper_id)
+ getOperDestinations(oper_id)
+ getOperatorProperties(oper_id)
+ getFirstOperator()
+ isLastOperator(oper_id)
+ getOperatorTime(oper_id)
+ getDegradationFactor()
+ accumBusyTime(oper_id, time)
+ accumInRecords(oper_id, n_records)
+ accumOutRecords(oper_id, n_records)
ResourceManager
- registered_resources
+ registerResource(taskManager)
+ assignResource(oper_id, node_id)
+ slotFrom(location)
Responsible for managing the execution of a single JobGraph.
Receives the JobGraph and is responsible for planning and coordinating the execution of the job.
JobMaster
- operator_locations
- job_graph
JobMaster(cluster_cfg,
+ addLocation(oper_id,
+ getLocations(oper_id)
+ operDestinations(oper_id)
OperatorLocation
+ node_id
+ slot_id
+ mssg_id
ActiveSubtask
+ subtask
+ core_id: un
+ idx_priority
ExecutionBuffer
- active_subtask
- prepared_subtasks
- head_index
ExecutionBuffer(capacity)
+ activateSubtask(core, idx_prior)
+ getActiveSubtask()
+ emplace(subtask)
+ front()
+ pop()
+ size()
+ empty()
Subtask
+ mssge_id
+ slot_id
+ lapse
DEVS Atomic Models
Node
TIME
id
processing
taskManager
jobManager
- internal_pendings
- unif_distribution
- nextId
Node(jobManager, n_cores)
+ id()
+ getTaskManager()
checkExternalTransitionFromSwitch(bag_inport)
searchNextOperatorDestinations(bag_outport)
DEVS specifications:
+ external_transition()
+ internal_transition()
+ confluence_transition()
+ output()
+ time_advance()
+ input_ports:
tuple<in_port<OperatorLocation>, ..>
+ output_ports:
tuple<out_port<OperatorLocation>, ..>
+ state
Execute the actual processing work in Flink.
The number of TaskSlots in a TaskManager determines the number of concurrent processing tasks (threads) that can be executed.
TaskManager
- execution_buffers
- taskslots
- active_subtasks
- slots_used
- n_cores
- next_slot_id
- pending_execs
- SUBTASKS_CAPACITY
- TASKSLOTS_CAPACITY
TaskManager(n_cores)
+ reserveSlot(oper_id)
+ getSlot(slot_id)
+ getSlots()
+ scheduleExec(mssg_id, slot_id, jobman)
+ checkQueuedExecution(slot_id, jobman)
+ getActiveExecutions()
+ terminateActiveExecutions()
+ pendingExecutions()
+ executing()
+ getExecutionBuffers()
- createSubTask(jobman, tasklot, mssg_id, slot_id)
NodeMaster
- extern_locations
- mustSendInmediatlyToSwitch()
- checkExternalTransitionfromProducer(bag_inport)
<<interface>>
Generator
VALUE, TIME
+ getPeriod()
+ output_message(): VALUE
Producer
- expo_distr
- requeriments
- rate
- arrivalRates
- arrival_time
- current_time
- current_rate
- next_rate
- generating
Productor(metricLogger)
Switch
- port_buffers
- transmitting
- send_time
- uniform_distr
- send()
MessageBuffer
+ messages
+ active
Each TaskSlot represents a fixed subset of the resources available in a TaskManager.
TaskSlot
- buffer
- operator_id
- using
- active
- head_index
TaskSlot(oper_id)
+ getOperator()
+ pushTuple(mssg_id)
+ popTuple()
+ nTuples()
+ isUsing()
+ setUsing(state)
+ isActive()
+ seActive(state)
+ pendingTuples()
MetricLogger
- metric_directory
- cluster_cfg
- system_metrics
- operator_metrics
- captures
- THROUGHPUT_BASE_PATH
- UTILIZATION_BASE_PATH
- MAX_NUM_METRICS
- PS_TO_SECOND
MetricLogger(cluster_cfg, metric_dir)
+ captureMetrics(rate: elapsed_t, process_req)
+ printMetrics()
+ logDynamicMetrics()
+ logMetrics()
+ getClusterCFG()
SysMetric
+ processed_req
+ elapsed_time
+ throughput
OperMetric
+ p_level
+ in_records
+ out_records
+ accum_busy_time
+ busy_time
+ utilization